INCH-POUND

MIL-S-71079 (AR)
30 April 1993

MILITARY SPECIFICATION

SOLENOID VALVE, 3-WAY: 12561784

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the 3-way solenoid valve used in the Automatic Fire Control System (AFCS) for the M109A6 self-propelled howitzer (SPH) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-795	Cushioning Material, Packaging (Flexible Closed Cell Plastic Film for Long Shipping Cycle Applications)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820

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MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-117	-	Bag, Sleeve, and Tubing - Interior Packaging
MIL-C-5501	-	Caps and Plugs, Protective and Moisture Seal, General, Specification for
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Operation
MIL-F-13927	-	Fungus Resistance Test Automotive Components
MIL-H-46170	-	Hydraulic Fluid, Rust Inhibited, Fire Resistant Synthetic Hydrocarbon Base

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1246	-	Product Cleanliness and Contamination Control Program
MIL-STD-2073	-	DOD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12561784 - Solenoid Valve, 3-Way

(Copies of drawings, publications, and other Government documents required by the contractors in connection with specific

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acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General description. The 3-way solenoid valve is an electro-hydraulic device. The valve directs the flow of hydraulic fluid in response to an electrical input signal.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Supply fluid conditions. The 3-way solenoid valve shall meet the requirements specified when operated at the following fluid supply conditions.

a. Rated supply pressure: 1600 ± 50 psi. (For information only, the 3-way solenoid valve shall function for pressures ranging from 500 to 3000 psi.)

b. Fluid supply temperature: 0°F to 140°F.

c. Contamination concentration level shall be in accordance with MIL-STD-1246, level 200, as a maximum.

d. Hydraulic fluid shall be in accordance with MIL-H-6083 or equivalent (hydraulic fluid in accordance with MIL-H-46170 may be used for testing).

3.2.1.1.1 Supply fluid temperature extremes. The 3-way solenoid valve shall allow fluid to flow, but does not need to meet the performance requirements specified during exposure to supply fluid temperatures of -50°F to 0°F and +140°F to +250°F. The 3-way solenoid valve shall meet the requirements specified after exposure to the temperature ranges specified above.

3.2.1.2 Operating voltage. The 3-way solenoid valve shall operate over a voltage range from 18 to 30 Vdc.

3.2.1.2.1 Pull-in voltage. The voltage required to cause flow at the servo valve port shall be less than or equal to 15.5 Vdc.

3.2.1.2.2 Drop-out voltage. With an energized solenoid valve, flow shall cease at the servo valve port if the control voltage is decreased to 0.5 Vdc or less.

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3.2.1.3 Pressure drop.

3.2.1.3.1 Actuated. With the 3-way solenoid valve in the actuated position, the pressure drop across the inlet and outlet ports shall not exceed 110 psi at a flow rate of 11 ± 0.5 gallons per minute (gpm).

3.2.1.3.2 Deactuated. With the 3-way solenoid valve in the deactuated position, the pressure drop across the inlet and outlet ports shall not exceed 110 psi at a flow rate of 11 ± 0.5 gallons per minute (gpm).

3.2.1.4 Internal leakage.

3.2.1.4.1 Actuated. With the 3-way solenoid valve in the actuated position, the leakage measured at the power mode port shall not exceed 100cc per minute for any inlet pressure up to 3000 psi.

3.2.1.4.2 Deactuated. With the 3-way solenoid valve in the deactuated position, the leakage measured at the servo valve port shall not exceed 100cc per minute for any inlet pressure up to 3000 psi.

3.2.1.5 Coil resistance. Coil resistance shall be 15 ohms minimum.

3.2.1.6 Actuation time. With the supply pressure applied to the pressure port and upon application of 24.0 ± 0.5 Vdc to the solenoid, the 3-way solenoid valve shall activate from the power mode port to the servo valve port in 0.150 second maximum. The 3-way solenoid valve shall deactivate from the servo valve port to the power mode port within 0.150 second maximum after removal of the voltage.

3.2.1.7 Operating current. The 3-way solenoid valve operating current shall not exceed 2.0 amperes at 30 Vdc supply voltage.

3.2.1.8 Insulation resistance. The insulation resistance of the assembly shall be no less than 10 megohms measured between any terminal of the electrical connector and the housing when measured in accordance with MIL-STD-202, Method 302, Condition B.

3.2.1.9 External leakage. There shall be no external leakage (see 6.8.1) with the 3-way solenoid valve subjected to the proof pressure conditions specified in 3.2.1.10.

3.2.1.10 Proof pressure.

3.2.1.10.1 Deactuated. The 3-way solenoid valve shall withstand a proof pressure of 4500 – 4600 psi applied to the

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pressure port for a period of 5 minutes. The servo valve port and power mode port shall be connected together with the 3-way solenoid valve deactuated. There shall be no external leakage or any evidence of deformation. No degradation of performance shall result from this test.

3.2.1.10.2 Actuated. The 3-way solenoid valve shall withstand a proof pressure of 4500 - 4600 psi applied to the pressure port for a period of 5 minutes. The servo valve port and power mode port shall be connected together with the 3-way solenoid valve actuated. There shall be no external leakage or any evidence of deformation. No degradation of performance shall result from this test.

3.2.1.11 Burst pressure. The 3-way solenoid valve shall not rupture with a burst pressure of 7500 - 7600 psi applied to the pressure port for a period of 5 minutes. The servo valve port and power mode port shall be connected together with the 3-way solenoid valve deactuated. The valve is not expected to work after this test.

3.2.2 Environmental.

3.2.2.1 Operational environment.

3.2.2.1.1 Ambient operating temperature. The 3-way solenoid valve shall meet the performance requirements in 3.2.1 while the operating temperature of the ambient air is within the range of -50°F to +140°F.

3.2.2.1.2 Humidity. The valve shall meet the performance requirements in 3.2.1 at a relative humidity up to a maximum of 100 percent across the temperature range defined in 3.2.2.1.1.

3.2.2.1.3 Salt fog. The valve shall meet the performance requirements in 3.2.1 during and after exposure to salt fog solution at a temperature between 90°F and 95°F. The solution is defined as 5 percent Sodium Chloride and 95 percent distilled water by weight. Fog density is approximately 1.057 quarts of salt solution per 264 gallons of chamber volume per 24 hours.

3.2.2.1.4 Vibration. The valve shall meet the performance requirements in 3.2.1 during and after being subjected to vibration as shown in table I using the axis configurations as shown in figure 1.

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TABLE I. Vibration

Test Phase	5-500 Hz Floor Level (g^2/Hz)	No. Sweeps	Narrowband 1 BW (Hz)	Narrowband 1 Ampl (g^2/Hz)	Narrowband 1 Sweep BW (Hz)	Narrowband 2 BW (Hz)	Narrowband 2 Ampl (g^2/Hz)	Narrowband 2 Sweep BW (Hz)	Narrowband 3 BW (Hz)	Narrowband 3 Ampl (g^2/Hz)	Narrowband 3 Sweep BW (Hz)	Narrowband 4 BW (Hz)	Narrowband 4 Ampl (g^2/Hz)	Narrowband 4 Sweep BW (Hz)	Narrowband 5 BW (Hz)	Narrowband 5 Ampl (g^2/Hz)
VERTICAL AXIS (11 minutes per test phase)																
V01	0.0010	6	18-30	0.0082	6	36-60	0.0081	12	———	———	—	———	———	—	———	———
V02	0.0024	6	36-42	0.4914	3	72-84	0.0391	6	108-126	0.0076	9	144-168	0.0275	12	———	———
V03	0.0026	2	48-60	0.2467	6	96-120	0.0319	12	144-180	0.0062	18	192-240	0.0135	24	———	———
V04	0.0034	2	66-96	0.1221	15	132-192	0.0249	30	198-288	0.0074	45	———	———	—	———	———
V05	0.0024	2	102-126	0.0307	12	204-252	0.0120	24	408-500	0.0079	48	———	———	—	———	———
TRANSVERSE AXIS (11 minutes per test phase)																
T01	0.0022	4	18-36	0.0585	9	36-72	0.0064	18	———	———	—	———	———	—	———	———
T02	0.0034	4	42-48	0.1639	3	84-96	0.0435	6	126-144	0.0128	9	168-192	0.0130	12	210-240	0.0239
T03	0.0032	2	54-66	0.0332	6	108-132	0.0223	12	162-198	0.0150	18	216-264	0.0179	24	270-330	0.0152
T04	0.0040	2	72-96	0.0457	12	144-192	0.0221	24	216-288	0.0660	36	288-384	0.0076	48	———	———
T05	0.0030	2	102-126	0.0024	12	204-252	0.0647	24	306-378	0.0072	36	408-500	0.0074	48	———	———
LONGITUDINAL AXIS (11 minutes per test phase)																
L01	0.0012	4	18-36	0.0069	9	36-72	0.0046	18	———	———	—	———	———	—	———	———
L02	0.0017	4	42-48	0.0305	3	84-96	0.0143	6	126-144	0.0040	9	168-192	0.0246	12	210-240	0.0037
L03	0.0022	2	54-66	0.0679	6	108-132	0.0034	12	162-198	0.0067	18	216-264	0.0032	24	———	———
L04	0.0030	2	72-90	0.0369	9	144-180	0.0229	18	216-270	0.0040	27	288-360	0.0052	36	360-450	0.0069
L05	0.0022	2	96-126	0.0157	15	192-252	0.0091	30	288-378	0.0076	45	384-500	0.0072	60	———	———

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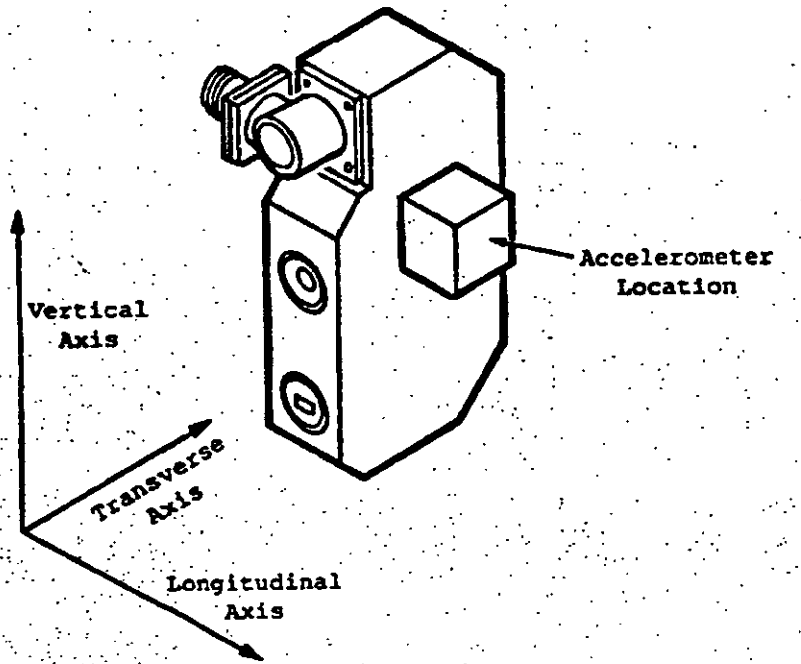


FIGURE 1. 3-way solenoid valve, axes orientations.

3.2.2.1.5 Shock.

3.2.2.1.5.1 Basic shock. The valve shall meet the performance requirements in 3.2.1 during and after being subjected to three half sine wave pulses applied along each of three mutually perpendicular axes with peak amplitudes of 30 ± 3 g for 18 ± 1.8 msec as shown in figure 1.

3.2.2.1.5.2 Gun firing shock. The valve shall meet the performance requirements in 3.2.1 during and after being subjected to three half sine wave pulses applied along each of three mutually perpendicular axes with peak amplitudes of 200 ± 20 g for 0.5 ± 0.1 msec as shown in figure 1.

3.2.2.1.6 Altitude. The valve shall meet the performance requirements in 3.2.1 during and after exposure to pressure encountered at altitudes up to 10,000 feet above sea level (10.1 psia).

3.2.2.2 Nonoperational environment.

3.2.2.2.1 Ambient storage temperature. The valve shall meet the performance requirements in 3.2.1 after being subject to storage temperature of the ambient air within the range of -50°F to +160°F per the following restrictions:

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a. Exposure at temperature extremes of 145°F to 160°F shall be limited to one hour duration per 24-hour period.

b. Temperature variation during any 8-hour period shall be sinusoidal with a maximum variation of 69°F.

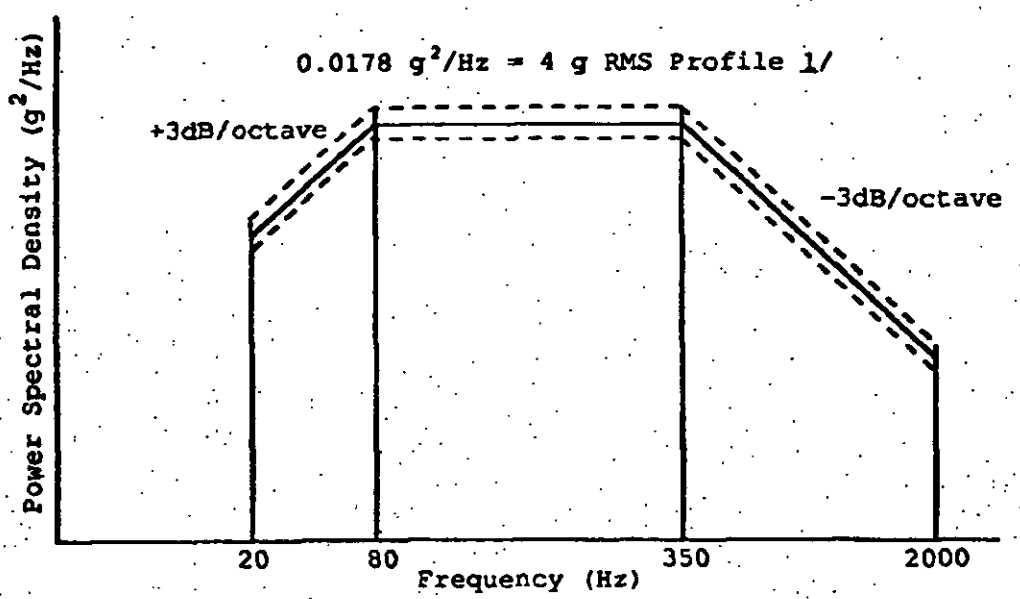
3.2.2.2.2 Fungus. The valve shall meet the performance requirements in 3.2.1 after exposure to spore suspension as defined in the "Specimen Inoculation" paragraph of MIL-F-13927, followed by exposure to ambient air temperatures between 81°F and 93°F at relative humidity between 96 and 100 percent, noncondensing for a 28 day duration. All exterior connectors and mounting surfaces shall have a mating connector or equivalent installed prior to exposure.

3.2.2.2.3 Altitude. The valve shall meet the performance requirements in 3.2.1 after exposure to pressures within the range of 1.68 to 2.14 psia.

3.2.2.2.4 Washdown. The valve shall meet the performance requirements in 3.2.1 after being subjected to washdown using hot water at 140°F ± 20°F with a shearing pressure between 9 and 11 psi and air drying at a temperature not to exceed 145°F. All exterior connectors and mounting surfaces shall have a mating connector or equivalent installed prior to exposure.

3.3 Environmental stress screening (ESS). Unless otherwise specified, the 3-way valve shall be subjected to random vibration per figure 2 in each of three axes for 10 minutes per axis (see 6.2). The valve shall be filled with oil prior to vibration. The average vibration level shall be monitored at the 3-way valve mounting points and shall be controlled using a control bandwidth of no greater than 10 Hz for vibration frequencies above 500 Hz. Instantaneous acceleration peaks shall be limited to three times the Root-Mean-Square (RMS) acceleration value. Following vibration, the 3-way valve shall meet the performance requirements in 3.2.1. All failures which occur as a result of screening shall be corrected and vibration repeated. If the 3-way valve is subjected to vibration three times before completing ESS, it shall be subjected to formal failure analysis.

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1/ Dashed line indicates tolerance band

FIGURE 2. Random vibration spectrum.

3.4 Fabrication. The valve shall be manufactured in accordance with drawing 12561784.

3.5 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.5.

3.5.1 First article sample inspection. After completion of first article test, the first article sample shall meet the requirements of 3.6.1.

3.6 Workmanship.

3.6.1 General workmanship. General workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the

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specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.5)
b. Quality Conformance (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions in 4.7.1.

4.4 Inspection equipment. Contractor developed special test equipment including test software required to perform examinations and tests prescribed herein shall be submitted for approval in accordance with the terms of the contract (see 6.2 and 6.4).

4.5 First article inspection.

4.5.1 Submission. When required by the procuring activity (see 6.2), the contractor shall submit a First Article sample for evaluation in accordance with the provisions of 4.5.2 and 4.5.3. The First Article sample size shall be three (3) or as specified in the contract or purchase order.

4.5.2 Inspections to be performed. The sequence of first article testing shall be in accordance with table II. The first article sample shall have been subjected to, and pass, the quality conformance requirements as defined in table III. Subassemblies and piece parts, at a minimum, shall be inspected for conformance to the characteristics defined in the quality assurance provisions of the applicable drawing.

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TABLE II. First article test sequence.

Test Sequence 1/	Requirement Paragraph	Test Method	Sample 1	2	3
Performance 2/	3.2.1	4.7.2	X	X	X
Cold ambient storage temperature	3.2.2.2.1	4.7.3.2.1.1	X	X	X
Hot ambient storage temperature	3.2.2.2.1	4.7.3.2.1.2	X	X	X
Cold ambient operating temperature	3.2.2.1.1	4.7.3.1.1.1	X	X	X
Hot ambient operating temperature	3.2.2.1.1	4.7.3.1.1.2	X	X	X
Altitude nonoperating	3.2.2.2.3	4.7.3.2.3	X	X	X
Altitude operating	3.2.2.1.6	4.7.3.1.6	X	X	X
Vibration	3.2.2.1.4	4.7.3.1.4	X	X	X
Basic shock	3.2.2.1.5.1	4.7.3.1.5.1	X	X	X
Gun firing shock	3.2.2.1.5.2	4.7.3.1.5.2	X	X	X
Washdown	3.2.2.2.4	4.7.3.2.4	X	X	X
Humidity	3.2.2.1.2	4.7.3.1.2	X		
Fungus	3.2.2.2.2	4.7.3.2.2		X	
Salt fog	3.2.2.1.3	4.7.3.1.3			X
First article sample inspection	3.5.1	4.7.3.3	X	X	X

1/ To ensure the First Article sample has been subjected to, and passed, the quality conformance tests as defined in table III the contractors records may be reviewed to assess compliance.

2/ The first article performance tests, as defined in table III, shall be performed only once. Performance testing, after the first article sample has been subjected to the various environments shall consist of the performance test defined in the quality conformance section of table III.

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TABLE III. Category of inspection.

Class.	Description	Requirements	Method 1/	First Article	Quality Conformance
	Performance	3.2.1	4.7.2		
Major	Operating voltage	3.2.1.2	4.7.2.1	X	100%
Major	Pull-in voltage	3.2.1.2.1	4.7.2.1.1	X	100%
Major	Drop-out voltage	3.2.1.2.2	4.7.2.1.2	X	100%
	Pressure drop	3.2.1.3	4.7.2.2		
Major	Actuated	3.2.1.3.1	4.7.2.2	X	100%
Major	Deactuated	3.2.1.3.2	4.7.2.2	X	100%
Major	Internal leakage	3.2.1.4	4.7.2.3	X	100%
Major	Actuated	3.2.1.4.1	4.7.2.3	X	100%
Major	Deactuated	3.2.1.4.2	4.7.2.3	X	100%
Major	Coil resistance	3.2.1.5	4.7.2.4	X	100%
Major	Actuation time	3.2.1.6	4.7.2.5	X	100%
Major	Operating current	3.2.1.7	4.7.2.6	X	100%
Major	Insulation resistance	3.2.1.8	4.7.2.7	X	100%
Major	External leakage	3.2.1.9	4.7.2.8	X	100%
	Proof pressure	3.2.1.10	4.7.2.9		
Major	Deactuated	3.2.1.10.1	4.7.2.9.1	X	100%
Major	Actuated	3.2.1.10.2	4.7.2.9.2	X	100%
Major	Burst pressure	3.2.1.11	4.7.2.10	X	
	Environmental tests	3.2.2	4.7.3		
	Operational environ.	3.2.2.1	4.7.3.1		
	Ambient oper. temp.	3.2.2.1.1	4.7.3.1.1		
Major	Cold	3.2.2.1.1	4.7.3.1.1.1	X	
Major	Hot	3.2.2.1.1	4.7.3.1.1.2	X	
Major	Humidity	3.2.2.1.2	4.7.3.1.2	X	
Major	Salt fog	3.2.2.1.3	4.7.3.1.3	X	
Major	Vibration	3.2.2.1.4	4.7.3.1.4	X	
	Shock	3.2.2.1.5	4.7.3.1.5		
Major	Basic shock	3.2.2.1.5.1	4.7.3.1.5.1	X	
Major	Gun firing shock	3.2.2.1.5.2	4.7.3.1.5.2	X	
Major	Altitude	3.2.2.1.6	4.7.3.1.6	X	
	Non-operational environ.	3.2.2.2	4.7.3.2		
	Ambient oper. temp.	3.2.2.2.1	4.7.3.2.1		
Major	Cold	3.2.2.2.1	4.7.3.2.1.1	X	
Major	Hot	3.2.2.2.1	4.7.3.2.1.2	X	
Major	Fungus	3.2.2.2.2	4.7.3.2.2	X	
Major	Altitude	3.2.2.2.3	4.7.3.2.3	X	
Major	Washdown	3.2.2.1.4	4.7.3.2.4	X	
Minor	First article sample inspection	3.5.1	4.7.3.3	X	
Major	ESS	3.3	4.7.4	X	100%
	Workmanship	3.6	4.7.5		
Minor	General workmanship	3.6.1	4.7.5.1	X	2/
Major	Solder workmanship	3.6.2	4.7.5.2	X	100%
Minor	Packaging	5	4.7.6		2/

1/ Unless otherwise specified in 4.7, sequence of functional tests and physical characteristic examinations are optional.

2/ Sample size is as defined by level II of table IV.

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TABLE IV. Attributes sampling inspection.

Inspection levels 1/		
Lot Size	II	IV
2 to 8	*	*
9 to 15	*	13
16 to 25	*	13
26 to 50	*	13
51 to 90	*	13
91 to 150	125	13
151 to 28	125	32

1/ Numbers under inspection levels indicate sample size: asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

4.5.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements. Disposition of the item shall be as specified in the contract or purchase order (see 6.3).

4.6 Quality conformance inspections.

4.6.1 Examinations and tests.

4.6.1.1 Classification of characteristics. Quality conformance examination and test shall be as specified in table III. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Subassemblies and piece parts shall be inspected for conformance to the characteristics defined in the quality assurance provisions of the applicable drawing.

4.6.1.2 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer

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for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.6.2 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the item shall be rejected. Disposition of the item shall be as specified in the contract or purchase order. (See 6.2.).

4.7 Methods of inspection.

4.7.1 Test conditions.

4.7.1.1 Environmental. Unless otherwise specified, all tests shall be performed at an atmospheric pressure between 28 and 32 inches of mercury, a temperature of 75° ± 25°F, and a relative humidity between 10 and 90 percent.

4.7.1.2 Test software. Test software includes supporting test software which resides in external test computer(s). Test software used for first article and quality conformance tests shall be submitted for approval (see 4.4) as part of the inspection test equipment.

4.7.1.3 Supply fluid conditions. Unless otherwise specified, all tests shall be performed with fluid at 100° ± 30°F. Hydraulic fluid shall be as specified in 3.2.I.1. Unless otherwise specified, the 3-way solenoid valve shall be operated at the supply pressure specified in 3.2.1.1.

4.7.1.4 Operating conditions. Unless otherwise specified, the input voltage applied to the connector pins to actuate the 3-way solenoid valve shall be 24 ± 0.5 Vdc and the input voltage applied to deactuate the 3-way solenoid valve shall be 0 +0.5/-0.0 Vdc.

4.7.1.5 Pressure tests sequence. Proof pressure test (see 4.7.2.9) and related tests shall be conducted prior to other tests. Burst pressure test (see 4.7.2.10) shall be conducted after all other testing has been completed.

4.7.1.6 Internal cleanliness. Prior to any performance testing, the internal cleanliness of the item shall be established by providing input fluid rated at the cleanliness level specified in 3.2.2.2, then establishing flow between each of the ports designated. This flow shall be maintained until the effluent meets level 200 of MIL-STD-1246. At the conclusion of acceptance

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testing, the 3-way solenoid valve internal cleanliness shall be reestablished by performing the same flushing and effluent check until the effluent meets level 200 of MIL-STD-1246. Once level 200 has been achieved in the unit, all open ports shall be plugged in accordance with MIL-C-5501, and the unit shall be sealed with a noncontaminating tamper-proof seal and adequately tagged and identified before being subjected to preparation for delivery or storage processing. The 3-way solenoid valve shall be visually inspected to verify it is provided with the cleanliness protection specified.

4.7.1.7 Pre-test oil filling. Unless otherwise specified herein, the valves will be oil filled prior to testing.

4.7.2 Performance test methods.

4.7.2.1 Operating voltage. The input voltage of the 3-way solenoid valve shall be set to 18 +0.5/-0.0 Vdc. Verify there is fluid flow from the servo valve port. Increase the input voltage to 30 - 0.5 Vdc. Verify there is fluid flow from the servo valve port. The operating voltage shall meet the requirements of 3.2.1.2.

4.7.2.1.1 Pull in voltage. The pressure at the pressure port shall be set to 500 +100/-0.0 psi. The voltage to the solenoid shall be slowly increased until the valve opens as evidenced by a flow at the servo valve port. The pull-in voltage measured shall meet the requirements of 3.2.1.2.1.

4.7.2.1.2 Drop out voltage. The pressure at the pressure port shall be set to 3000 +100/-0.0 psi. Energize the valve with 30 +0.0/-0.5 Vdc. Slowly reduce the voltage until the valve closes as seen by stoppage of flow at the servo valve port. The voltage measured shall meet the requirements of 3.2.1.2.2.

4.7.2.2 Pressure drop. With the 3-way solenoid valve in the actuated position, the pressure drop across the inlet and outlet ports shall be measured. Apply rated pressure to the pressure port and restrict the flow from the servo valve port to 11 ± 0.5 gpm. Measure and record the fluid pressure exiting from the servo valve port. Deenergize the valve and adjust flow from the power mode port to 11 ± 0.5 gpm. Measure and record the fluid pressure exiting the power mode port. The pressure drop shall meet the requirements of paragraph 3.2.1.3.

4.7.2.3 Internal leakage. With the 3-way solenoid valve in the deactuated position, apply 3000 +100/-0 psi to the pressure port (reference figure 3). Hold this pressure for five minutes minimum. Observe the leakage at the servo valve port. The leakage measured at the servo valve port shall meet the requirements of paragraph 3.2.1.4. Repeat this procedure with the

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valve actuated and measure leakage at the power mode port. The measured internal leakage shall meet the requirements of paragraph 3.2.1.4.

4.7.2.4 Coil resistance. The coil resistance shall be measured using an ohmmeter with a minimum of 1.0 megohm input impedance and shall meet the requirements of 3.2.1.5.

4.7.2.5 Actuation time. With the supply pressure applied to the pressure port and upon application of 24 ± 0.5 Vdc to the solenoid, the actuation time of the 3-way solenoid valve shall be measured and shall meet the requirements of paragraph 3.2.1.6. Upon removal of the supply voltage, the deactivation time of the valve shall meet the requirements of 3.2.1.6.

4.7.2.6 Operating current. The solenoid input voltage shall be set to 30 +0.0/-0.5 Vdc and solenoid operating current shall be measured and shall meet the requirements of 3.2.1.7.

4.7.2.7 Insulation resistance. The 3-way solenoid valve shall be tested for insulation resistance in accordance with MIL-STD-202, method 302, test condition B and shall meet the requirements of 3.2.1.8.

4.7.2.8 External leakage. The external leakage shall meet the requirements of 3.2.1.9 when the 3-way solenoid valve is subjected to the proof pressure tests in 4.7.2.9.

4.7.2.9 Proof pressure. Fluid flow from the servo valve and power mode ports shall be blocked. Figure 3 shows the proof pressure test setup.

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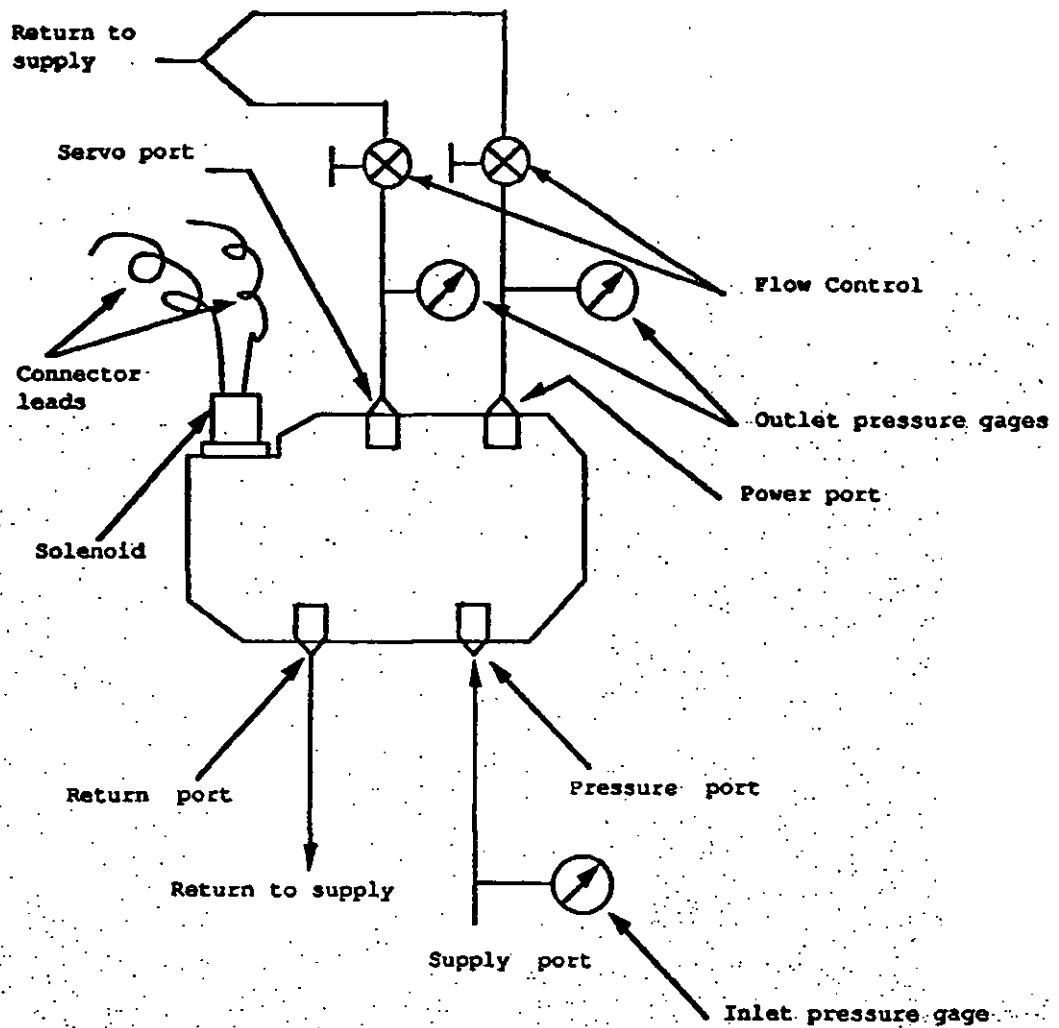


FIGURE 3. Test setup for proof pressure and leakage tests.

4.7.2.9.1 Deactuated. The solenoid shall be deactuated and pressure to the pressure port slowly brought up to between 4500 and 4600 psi. Hold this pressure for five minutes minimum and observe for external leakage and structural deformation or rupture. The 3-way solenoid valve shall meet the requirements of 3.2.1.10.1. Reduce pressure.

4.7.2.9.2 Actuated. The solenoid shall be actuated and the pressure to the pressure port slowly (at a rate less than 400 psi/sec) brought up to 4500 to 4600 psi. Hold this pressure for five minutes minimum and observe for external leakage and

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structural deformation or rupture. The 3-way solenoid valve shall meet the requirements of 3.2.1.10.2. Reduce pressure.

4.7.2.10 Burst pressure. Fluid flow from the servo valve and power mode ports shall be blocked. Figure 3 shows a possible burst pressure test setup. The solenoid shall be deactuated and pressure to the pressure port, servo valve port, and power mode port slowly (at a rate less than 400 psi/sec) brought up to 7500 + 100 psi so that all ports are at this pressure. Maintain this pressure on all ports for a period of 5 minutes +0/-30 seconds. Observe for structural deformation or rupture. The 3-way solenoid valve shall meet the requirements of 3.2.1.11. Reduce pressure. The valve assembly shall not rupture as a result of this test.

NOTE: The valve assembly, subjected to burst pressure, shall not be shipped with the regular contract or order and shall be indelibly marked "DO NOT USE."

4.7.3 Environmental test methods.

4.7.3.1 Operational environment.

4.7.3.1.1 Ambient operating temperature.

4.7.3.1.1.1 Cold ambient operating temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 502.2, Procedure II after thermally stabilizing at a low temperature of -50° +5°/-0°F for a minimum of 4 hours. During the stabilization period, the unit shall be unpowered. At the conclusion of this time, the 3-way solenoid valve shall be tested per the quality conformance performance tests specified in table III. If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the temperature conditions shall be re-established prior to continuing.

4.7.3.1.1.2 Hot ambient operating temperature. This test shall be conducted in accordance with and MIL-STD-810D, Method 501.2, Procedure II after thermally stabilizing at a high temperature of +140° +0/-5°F for a minimum of 4 hours. During the stabilization period, the unit shall be unpowered. At the conclusion of this time, the valve shall be tested per the quality conformance performance tests specified in table III. If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the temperature conditions shall be re-established prior to continuing.

4.7.3.1.2 Humidity. This test shall be conducted in accordance with MIL-STD-810D, Method 507.2, Procedure I, Cycle 2,

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for 240 hours (10 cycles) except that the temperature shall be cycled from 85°F to 140°F during each 24-hour cycle. During each 24-hour cycle, the temperature shall be stabilized at 85°F for a minimum of four hours and at 140°F for a minimum of four hours. The rate of temperature change during transition periods shall not exceed 8°F/hour. All exterior connectors or mating surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. Operational checkouts shall be conducted using the quality conformance performance tests specified in table III within 1 hour of removal of the unit from the chamber and shall be performed at the end of the tenth cycle as a minimum.

4.7.3.1.3 Salt fog. This test shall be conducted in accordance with MIL-STD-810D, Method 509.2. The unit under test shall be unpowered while exposed to a salt fog solution of 5 percent ± 1 percent Sodium Chloride for 48 hours. The temperature in the exposure zone shall be maintained between 90°F and 95°F. All exterior connectors or mounting surfaces shall have a mating connector, cover, or equivalent installed during exposure. After exposure, the valve shall be dried for 48 hours and tested per the quality conformance performance tests specified in table III.

4.7.3.1.4 Vibration. This test shall be conducted in accordance with MIL-STD-810D, Method 514.3, Procedure I, Category 8. The vibration test levels and duration are specified in table I in 3.2.2.1.4. After exposure, the valve shall be tested per the quality conformance performance tests specified in table III. Location of the vibration control accelerometers shall be as shown in figure 1. The duration for each phase shall be as specified in table I plus or minus 30 seconds.

4.7.3.1.5 Shock.

4.7.3.1.5.1 Basic shock. This test shall be conducted in accordance with MIL-STD-810D, Method 516.3 Procedure I and shall consist of three cycles of shock as shown in 3.2.2.1.5.1. The response spectra shall not deviate from the input condition by more than ± 3.5 dB. After exposure, the valve shall be tested per the quality conformance performance tests specified in table III.

4.7.3.1.5.2 Gunfire shock. This test shall be conducted in accordance with MIL-STD-810D, Method 516.3 Procedure I and shall consist of three cycles of shock as shown in 3.2.2.1.5.2. The response spectra shall not deviate from the input condition by more than ± 3.5 dB. After exposure the valve shall be tested per the quality conformance performance tests specified in table III.

4.7.3.1.6 Altitude. This test shall be conducted in accordance with MIL-STD-810D, Method 500.2 Procedure II. Pressure shall be maintained at 10.1 psia for a minimum of 30 minutes. During the stabilization period, the unit shall be unpowered. At the conclusion of this time, the valve shall be tested per the quality conformance performance test specified in table III.

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4.7.3.2 Nonoperational environment.

4.7.3.2.1 Ambient storage temperature.

4.7.3.2.1.1 Cold ambient storage temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 502.2, Procedure I for 168 hours at a low temperature of -50° +5°/-0°F. After exposure, the valve shall be tested per the quality conformance performance tests specified in table III at -50° +5°/-0°F and again at room temperature (75° ± 25°F).

4.7.3.2.1.2 Hot ambient storage temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 501.2, Procedure I for 168 hours (7 cycles) at a high temperature of +145° +0°/-5°F, with 1 hour +0/-5 minutes of each 24 hour cycle at 160° +0°/-5°F per figure 4. After exposure, the valve shall be tested per the first article performance tests specified in table III at room temperature (75° ± 25°F).

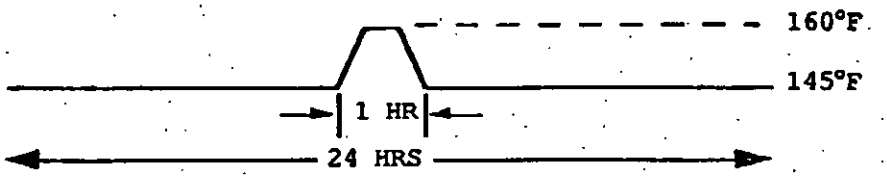


FIGURE 4. Hot storage temperature profile.

4.7.3.2.2 Fungus. This test shall be conducted in accordance with MIL-F-13927 Class 2, Method B except temperature and duration shall be as specified in 3.2.2.2. All exterior connectors or mounting surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. After exposure, the valve shall be tested per the quality conformance performance tests specified in table III at room temperature (75° ± 25°F).

4.7.3.2.3 Altitude. This test shall be conducted in accordance with MIL-STD-810D, Method 500.2, Procedure I and 3.2.2.2.3. Pressure shall be maintained at specified value for 4 hours -0/+5 minutes. All exterior connectors or mounting surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. After exposure, the valve shall be tested per the quality conformance performance tests specified in table III.

4.7.3.2.4 Washdown. The valve shall be subjected to a washdown in accordance with 3.2.2.2.4 for 5 minutes ± 1 minute uniformly applied over external surfaces and air dried for 2 hours ± 5 minutes at a temperature not to exceed 145°F. All exterior connectors and mounting surfaces shall have a mating connector, cover, or equivalent installed prior to exposure. After exposure

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and drying, the valve shall be tested per the quality conformance performance tests specified in table III.

4.7.3.3 First article sample inspection. The item shall be visually inspected (externally) for damage resulting in workmanship defects in accordance with 3.5.1. The cover shall be removed and the item interior shall be visually inspected for damage resulting in workmanship defects in accordance with MIL-STD-454, Requirement 9.

4.7.4 Environmental stress screening (ESS). The manufacturing process shall be monitored for compliance to 3.3.

4.7.5 Workmanship.

4.7.5.1 General workmanship. General workmanship shall be in accordance with MIL-STD-454, Requirement 9. The valve shall be visually inspected for legible identification markings specified in accordance with the applicable drawing.

4.7.6 Packaging inspection.

4.7.6.1 Submission. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product. Unless otherwise specified, sampling for examination shall be in accordance with 4.6.1.

4.7.6.2 Inspection to be performed.

4.7.6.2.1 Level A. When Level A packaging is specified in the contract, samples, selected in accordance with table III, shall be inspected for the defects identified in table V to verify compliance with section 5.

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TABLE V. Level A packaging inspection provisions.

Requirement	Inspection Criteria	Method
Cleaning	Item not completely clean. Item damaged by cleaning process.	Visual Visual
Drying	Item not completely dry. Item damaged by drying process.	Visual Visual
Unit packing	Bubble wrap incomplete or not secured. Improper amount of desiccant. Desiccant contacting item. Heat seal test. Humidity indicator not properly placed. Unit container not as specified. Cushion material not as specified. Improper closure of unit container. Excessive looseness in unit container.	Visual Visual Visual 1/ Visual Visual Visual Visual Shake/Visual
Marking	Marking illegible, incorrect, or incomplete.	Visual

1/ Test in accordance with MIL-P-116, group B inspection.

4.7.6.2.2 Level C. When Level C packaging is specified in the contract, samples, selected in accordance with table III, shall be inspected for the defects identified in table VI to verify compliance with section 5.

TABLE VI. Level C packaging inspection provisions.

Requirement	Inspection Criteria	Method
Unit packing	Unit container not as specified. Improper closure of unit container. Excessive looseness in unit container.	Visual Visual Shake/Visual
Marking	Marking illegible, incorrect, or incomplete.	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C as specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning. The assembly shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. The assembly shall be dried in accordance with MIL-P-116, as required.

5.1.1.3 Preservative application. Preservative shall not be applied.

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5.1.1.4 Unit packs. Each assembly shall be individually unit packed in accordance with submethod IIb of MIL-P-116 in a bag in accordance with MIL-B-117, type 1, class F, style 1. To prevent puncturing of the bag, any protruding parts or sharp edges shall be cushioned with cushion material, flexible, cellular plastic film, class 2 per PPP-C-795 (bubble pack) prior to placement in the bag.

5.1.2 Level C. Unit pack shall be as required to safely transport and handle the 3-way solenoid valve throughout the distribution cycle.

5.2 Packing. Unless otherwise specified by contract or purchase order, packing of the assembly shall be in accordance with Level A or C requirements of MIL-STD-2073.

5.3 Marking.

5.3.1 Standard marking. In addition to any special or other identification marking required by the contract, each unit and exterior container shall be marked in accordance with MIL-STD-129.

5.3.2 Special marking. In addition to the marking requirements of 5.3.1 and regardless of the level or type of packaging specified, special markings may also be required (see 6.2).

5.4 General.

5.4.1 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.7.6

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 3-way servo valve is intended for use as a component of the M109A6 Howitzer automatic fire control system.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Time frame required for submission of first article and number of units required (see 4.5).

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d. Levels of preservation and packaging (see 5.1 and 5.2).

e. Permission for the contractor to optimize or tailor ESS and preparation of an ESS optimization or tailoring plan per MIL-STD-785 Task 301, if applicable (see 3.3).

f. Special test equipment approval requirements (see 4.4).

g. Disposition instructions for nonconforming material (see 4.5.3 and 4.6.2).

h. Special marking requirements (see 5.3.2).

i. Pertinent first article information (see 6.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a preproduction sample, a first article sample, a first production item, and the number of items to be tested as specified in 4.5.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired and tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Submission of inspection equipment designs for approval. Contractor designs for final acceptance inspection equipment should be approved by the Government prior to fabrication or procuring the equipment. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1423 in the contract.

6.5 Subject term (key word) listing.

AFCS
Fire Control System, Automatic

6.6 Precautions. Prior to any system level usage or performance testing, the internal cleanliness of the item should be established by providing input fluid rated at the cleanliness level specified in 3.2.2.2, then establishing flow between each of the ports designated. This flow shall be maintained until the effluent meets level 200 of MIL-STD-1246.

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6.7 EMI/EMP/Nuclear

6.7.1 Electromagnetic radiation.

6.7.1.1 Electromagnetic interference/compatibility. The control of electromagnetic emission and susceptibility characteristics of the assembly shall be maintained in accordance with MIL-STD-461B. Electromagnetic compatibility requirements shall be maintained in accordance with MIL-E-6051D(1).

6.7.1.2 Electromagnetic radiation hazard effects. The assembly shall be maintained for safety and reliability in accordance with the Army Electromagnetic Radiation (EMR) criteria of the Adjutant General's Office (TAGO), listed in technical report MICOM-TR-RT-81-5.

6.7.1.3 Lightning effects. The assembly shall be maintained to be operationally unaffected after a near hit lightning strike (10 meters) when installed in an an M109A6 Self Propelled Howitzer. Lightning and electrostatic hazards shall be as specified in MICOM-TR-RT-81-5.

6.7.1.4 Electromagnetic pulse effects/nuclear survivability/ vulnerability. The assembly shall be maintained to meet an EMP environment, when installed in an an M109A6 Self Propelled Howitzer, as specified in the "Nuclear Survivability Criteria for Howitzer Improvement Program (HIP)," established by the U.S. Army Nuclear and Chemical Agency (USANCA) letter for MONA-NU to AMCPM-CAWS-W, Subject: Request for Nuclear Survivability Criteria, dated 15 Mar 85 (document CONFIDENTIAL). The assembly baseline survivability hardness levels are described in the reports:

a. Test Report, Preproduction Qualification Test (PQT) of Howitzer Improvement Program (U), Edwin K. Dunlap, Judith Ross, Nuclear Survivability Assessment Division, Nuclear Effects Directorate, White Sands, New Mexico, dated November 1989, TECOM Project No. 2-WE-200-HIP-029 (document SECRET).

b. Memorandum for Program Manager, Howitzer Improvement Program, ATTN: AMCPM-HIP, Dover, NJ, dated 14 August 1991. SUBJECT: Interim Report on Corrective Action Testing of M109A3E2 Howitzer (HI) Nuclear Survivability Assessment Program (NSA), TECOM Proj #2-WE-206-HIP-209 (U), Technical Risk Assessment, 6 Parts (U) (document SECRET).

6.8 Expanded definitions. The 3-way solenoid valve directs the flow of hydraulic fluid in response to an electrical input signal. The valve is in line with the Howitzer hydraulic system, and provides hydraulic pressure to either the elevation and azimuth servo valve assemblies or the manual hydraulic power handle (see figure 5). The Ballistic Computer/Weapon Controller (BC/WC) provides electrical control of the valve. If the valve is

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actuated by power applied to the coil, hydraulic pressure is applied to the elevation and azimuth servo valves. If the valve is deactuated, hydraulic pressure is applied to the manual hydraulic power handle. Figure 6 is an hydraulic schematic of the valve. Figure 7 is an electrical schematic of the 3-way solenoid valve.

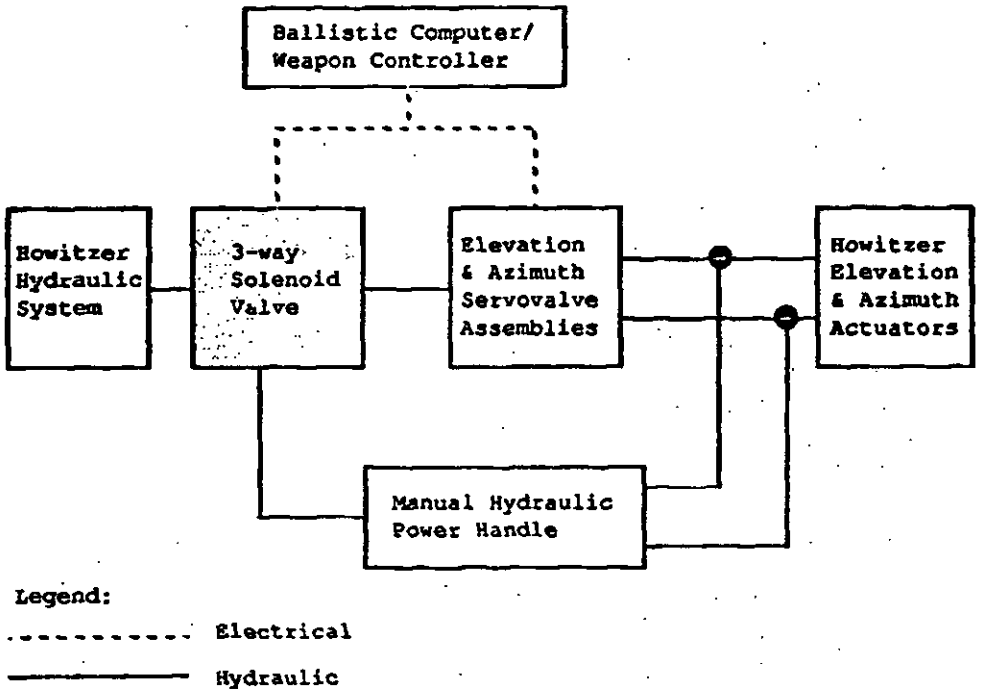


FIGURE 5. 3-way solenoid valve system interface.

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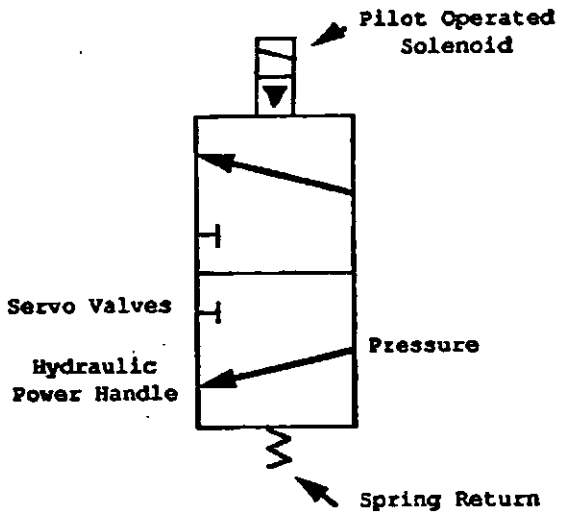


FIGURE 6. 3-way solenoid valve hydraulic schematic.

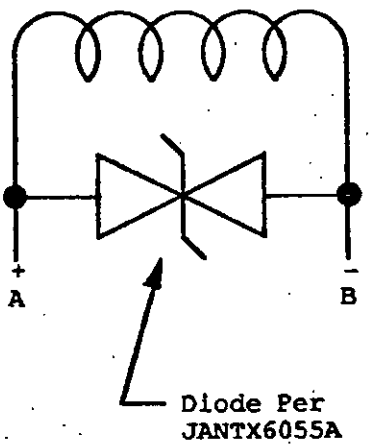


FIGURE 7. 3-way solenoid valve electrical schematic.

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